Muhammad Hulwan Afif Bin Rajali

HA

Field Data Processing Lead · Geospatial & LiDAR Specialist

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PROFILE

Geospatial and LiDAR professional with 8+ years across field operations, point-cloud processing, quality control and delivery leadership. Leads mobile and terrestrial LiDAR operations for NParks Singapore's national tree and road-network programme, covering 832,000+ trees and approximately 3,500 km of road. Experienced in UAV mapping, 3D scanning, semantic extraction, GIS and digital-twin workflows across Singapore, Malaysia, Brazil, Australia and the Europe region.

832,000+	~3,500 km	40%	5
TREES VALIDATED	ROAD NETWORK	WORKFLOW GAIN	COUNTRIES & REGIONS

PROFESSIONAL EXPERIENCE

Field Data Processing Lead / Delivery Manager

2023 – Present

GREEHILL ASIA-PACIFIC PTE LTD · SINGAPORE & APAC

- Lead LiDAR data operations across NParks Singapore and Arborise-linked managed areas; plan and monitor field and processing delivery against schedule and quality targets.
- Own semantic processing for automated Tree Semantic Extraction (TSE): classification, segmentation, validation and attribute completeness.
- Validate national datasets covering 832,000+ trees and approximately 3,500 km of road; resolve poor-GNSS and legacy-data issues that blocked delivery.
- Coordinate processing work linked to Brazil, Australia and the Europe region; improved overall workflow performance by 40% through process optimisation.
- Train APAC partners, clients and city-council stakeholders; strengthen collaboration between FDP, validation, TSE and engineering teams.

Surveyor / 3D Scanning & Geospatial Data Analyst

2018 – 2023

YJP SURVEYORS PTE LTD · SINGAPORE

- Planned and executed aerial, terrestrial and mobile LiDAR surveys for public and private clients in Singapore, including plan submissions and technical reporting.
- Delivered 3D scanning across reclamation, construction, forestry, coastline, refinery, façade, heritage and vessel volumetric work.
- Operated Trimble, Leica, RIEGL, NavVis and Faro systems; processed georeferenced point clouds into DTM/DSM, orthophoto, volumes, change detection and as-built deliverables.
- Built Excel macro automation for vessel volume calculation and GIS-linked reporting workflows.
- Supported A*STAR R&D in BVLOS UAV operations, 5G transmission, mixed-reality/HoloLens visualisation and Scan-to-BIM/digital-twin workflows.

Assistant Resident Engineer / GIS Executive

2017 – 2018

GEO STRATA SDN BHD · MALAYSIA

- Led field data collection and database generation for the MBSA Asset Management System across 56 sections covering 29,030 hectares.
- Delivered MBPJ Asset Management System work across 9,720 hectares, including LiDAR processing, database generation and GIS integration.
- Supervised slope improvement and retaining-wall works; conducted as-built and setting-out surveys across Selangor.
- Led GIS data migration and staff training for Selangor Town and Country Planning.

Assistant Surveyor / GIS Trainer

2016 – 2017

N ZECK SERVICE & SKY SHINE CORPORATION · MALAYSIA

- Setting-out and as-built surveys in Sungai Buloh and Cyberjaya; boundary work and detail survey preparation.
- Delivered ArcGIS staff training and supported migration from MapInfo to ArcGIS for JPBD spatial datasets.

Intern

2015

DEPARTMENT OF SURVEY AND MAPPING MALAYSIA (JUPEM) · MALAYSIA

- Supported title survey processes: registered survey applications, cadastral survey, CRM establishment and subdivision work.

SELECTED PROJECT EXPOSURE

NParks Singapore — national tree & road network	Mobile and terrestrial LiDAR delivery, semantic tree extraction and QC for 832,000+ trees and ~3,500 km of road.
Refinery & industrial as-built	Terrestrial scanning and registered point clouds for Shell, ExxonMobil, SRC and Vopak facilities.
MBSA & MBPJ asset management	Field collection and GIS databases across 38,750 hectares, including LiDAR processing and staff training.
A*STAR geospatial R&D	BVLOS UAV operations, 5G transmission and mixed-reality/HoloLens geospatial visualisation.

SELECTED CLIENTS & AGENCIES

NParks · JTC · PUB · MPA · HDB · SLA · NEA · NHB · Shell · ExxonMobil · SRC · Vopak · Resorts World Sentosa · A*STAR

TECHNICAL SKILLS

CAPTURE	Mobile and terrestrial LiDAR · UAV mapping · 3D laser scanning · GNSS, total station and setting-out · as-built and topographic survey
PROCESSING	RiProcess · POSPac · PhoenixLiDAR · Leica Cyclone and Infinity · Trimble Business Center · CloudCompare · LAStools
DATA & GIS	PostgreSQL · ArcGIS · QGIS · MapInfo · FME · DTM/DSM · orthophoto · change detection · volumetrics · Excel automation
DELIVERY	Geospatial QC and validation · TSE support · workflow optimisation · team coordination · partner training · client handover
R&D	BVLOS UAV · 5G transmission · AR/VR and HoloLens · Scan-to-BIM · digital twin

EDUCATION

BSc Surveying Science & Geomatics
First Class Honours · Universiti Teknologi MARA (UiTM) · 2014–2017

Diploma in Civil Engineering
Universiti Teknologi MARA (UiTM) · 2011–2014

Sijil Pelajaran Malaysia
Sekolah Menengah Teknik Port Dickson · 2009–2010

CERTIFICATIONS

Unmanned Aircraft Pilot Licence (UAPL)
Civil Aviation Authority of Singapore · 150 flight hours (Aug 2023)

Certified Drone Operator — Alta X
Freefly Systems Training Programme · Seattle, USA · 2023

DJI UAS Training
Aerial Mapping & Surveying · DJI Academy · 2023

WSH for Construction Work
Singapore Workforce Skills Qualifications · 2019

WSH in Process Plant
Singapore Workforce Skills Qualifications · 2020

PROFESSIONAL MEMBERSHIPS

- Professional Member, Royal Institution of Surveyors Malaysia (RISM)
- Professional Member, Institution of Geospatial and Remote Sensing Malaysia (IGRSM)
- Graduate Technologist, Malaysia Board of Technologists — Survey & Geomatics Technology, 2021